

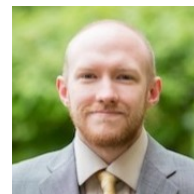
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Summary

Experienced data scientist and software engineer with 12 years spent designing, conducting, and sharing results of complex research. Worked with faculty, staff, and students from dozens of top universities to achieve excellence in the preparation and performance of research-related tasks, including data management, hypothesis formulation, and both mathematical and statistical modeling.

Possesses a diverse set of software skills, including DevOps, full-stack app development, performance benchmarking, ETL, and cloud deployments. Enjoys hosting workshops and hackathons to train users in state-of-the-art technologies and best practices. Familiar with a large number of data formats, with a strong concentration on updating local file standards to more cloud-performant ones. Knowledgeable about various mathematical and statistical applications, including high-dimensional time series analysis, nonlinear dynamics, and machine learning.

Education

- 2015 – 2020  **Ph.D.** | Applied and Computational Mathematics and Statistics | *University of Notre Dame*
Doctoral thesis: "Second-order moments of activity in large neural network models"
- Relevant coursework: machine learning, Bayesian statistics, network science, stochastic analysis, time series analysis, partial differential equations, nonlinear dynamics
- 2011 – 2015  **B.S.** | Applied Mathematics and Cognitive Science | *University of Evansville*
Minors in Computer Science, Philosophy, Psychology, and Neuroscience
Senior thesis: "Crime prediction models applied to the city of Evansville"
Honors thesis: "Goodness of fit metrics for joint probability distributions"
- Relevant coursework: calculus, linear algebra, probability, statistics, discrete & combinatorics, numerical analysis, real analysis, mathematical physics, algorithms & data structures, cryptography, machine learning, symbolic logic, cognitive psychology, neurophysiology, advanced neuroscience, philosophy of mind, philosophy of science

Experience

- 2025 – Present  **Research Software Engineer** | *Center for Open Neuroscience* | *Dartmouth College* | Hybrid
- Maintaining and enhancing several NIH data archives that store neuroscientific data by standardizing dataset organization across multiple modalities and providing usage analytics, including geolocation, through parsing terabytes of raw S3 log files.
 - Assisting with user support tickets and feature requests to ensure smooth operation of data submission and retrieval processes.
 - Developing open source software for use in neurophysiology research with an emphasis on achieving reproducibility of scientific findings.
 - Researching options for stable, cost-effective, long-term backup of petabyte-scale data storage. Working with a range of institutions and services to provide solutions and cost estimates to that effect.
 - Providing software, consultation, and computing support for the Brain Behavior Quantification and Synchronization consortium (BBQS).

Experience (continued)

2020 – 2024  **Research Software Engineer** | *CatalystNeuro* | Remote

- Developed and maintained 5 repositories of open-source software by ensuring proper functionality of automated testing suites, documentation, tutorials, and demos.
- Created and maintained 12+ data processing pipelines for neuroscience labs, allowing their data to flow from acquisition to sharing in a seamless fashion.
- Curated a total of 256 TB of high-value datasets to NIH data archives on behalf of various research groups.
- Managed the company's cloud resources (Amazon Web Services; AWS), including storage, compute resources, and identity access management (IAM).
- Handled user interactions within the research community by offering technical support and resolving issues or feature requests in a timely manner.
- Facilitated user education across various platforms by running sessions at multiple conferences and workshops, increasing user adoption and effective system utilization.
- All software related in some way to the facilitation of terabyte-scale data management, analysis, and visualization for the field of neurophysiology.

2016 – 2020  **Research Assistant** for the Neural Computing Group at *University of Notre Dame*

- Reconciled the computational properties of biologically realistic neural networks with artificial machine learning models (such as those used across computer vision) through complex mathematical theory and stochastic biophysical simulations, with results communicated through 3 journal publications and a presentation at the high-impact COSYNE conference (top 4% of abstracts accepted).
- Collaborated with several top experimentalists in neuroscience as a trainee in the NeuroNex program, which focused on understanding how neural function emerges from underlying structure.

2015 – 2017  **Teaching Assistant** for Applied Mathematical Methods at *University of Notre Dame*

- Ran 4 tutorial sections for 112 students, aiding their study of course concepts to achieve a 96% satisfaction rate.
- Graded homework assignments and exams. Filled in main lectures for the professor as needed.

2015 - 2016  **Data Analyst** for *JRM Environmental Inc*

- Analyzed geological data from water samples in conjunction with the Indiana Department of Environmental Management to issue compliance permits for Indianapolis regulation standards.

Summer 2015  **Technical Editor** for *Penguin Random House*

- Reviewed and corrected over 300 pages of "An Idiot's Guide to Algebra II".

Summer 2014  **Research Intern** for the BITLab at *Michigan State University*

- Explored statistical effects of algorithmic curation (the use of automated filtering mechanisms in the delivery and display of information) by measuring properties of simulated models of social networks, with results presented at the MIDSURE conference.

Summer 2013  **Research Intern** | Bioinformatics | *University of Kansas*

- Examined information diffusion through large-scale simulations of G-protein signaling mechanisms using a high-performance super-computing cluster (HPC), then presented results at an undergraduate symposium.

Key Achievements

Data	 Published over 256 TB of neuroscience data to NIH archives, allowing anyone in the world to test scientific hypotheses.
Education	 Trained 200+ professionals through 7 workshops and hackathons, disseminating the practices of open data.
Software	 Developed software used by more than 50 research groups globally, integrating with daily data pipelines and practices.
Standardization	 Standardized metadata of 40 data formats used by scientific devices, creating a universal language for comparing results.
Publications	 Published 7 papers and 8 conference posters, spreading the word about open-source software and public data.

Key Achievements (continued)

Pipelines  Updated **10** data analysis pipelines to leverage cloud streaming capabilities to improve speed and robustness.

Skills

Coding languages	 Python, MATLAB, Typescript, Javascript, HTML, LaTeX, R, C, C++, Java, CSS
Math, stats, & ML	 Multivariate regression, classification, cross-validation, clustering, mixture models, dimensionality reduction, ensemble methods, deep learning, reinforcement learning, time series analysis, anomaly detection, computer vision
DevOps	 Git, GitHub Actions, Docker, unit & integration testing, user & quality testing, Storybook, Chromatic, Puppeteer, Codecov
Libraries	 NumPy, SciPy, pandas, polars, sklearn, OpenCV, FFmpeg, timeit, airspeed velocity, Matplotlib, Plotly, Pydantic, Boto3, concurrents, joblib, tqdm, requests, pytest, Flask, PyInstaller
Frameworks & APIs	 AWS (IAM, S3, EC2, Batch, ECR, Route 53), Terraform, IPInfo, OpenCage, fsspec, HDF5, ROS3, remfile, h5wasm, obstore, Electron, Tk, Qt, React, pre-commit, Ruff, Black, flake8, pydocstyle, Mypy, Prettier, multiprocessing, threading, high-performance computing (HPC), Setuptools, Hatch, Twine, conda-forge, Vite, Node.js, Yarn
Databases	 REDIS, DynamoDB, MongoDB, PostgreSQL, SQLite, NoSQL
Data formats	 HDF5, Zarr, JSON, YAML, XML, Markdown, RST, Parquet, TSV, XLSX, MP4, AVI, TIFF, PNG, JPEG, MP3, WAV, PKL, MAT
Administration	 GitHub, AWS, PyPI, Condaforge, DockerHub, ReadTheDocs
Business & team skills	 Agile, scrum, research, mentoring, training, client consultations, drafting scopes of work, budgeting, grant writing

Highlighted Projects

Below are the most recent highlights of my major work. You can find the full listing at codycbakerphd.com/#highlighted-projects

NWB GUIDE | *Lead Developer*

- Developed an intuitive user interface for file management using interactive validation and real-time suggestions, streamlining the process for data submission to NIH archives.
- Ensured a robust testing suite involving multiple levels of integration and user interactions emulated using Puppeteer to enhance reliability and long-term maintainability.
- Tracked and documented dozens of hands-on user tests to refine the user experience.

nwb2bids | *Lead Developer*

- Created a metadata extraction tool for organizing datasets of NeurodataWithoutBorders (NWB) files following the Brain Imaging Data Structure (BIDS) structure.
- Encourages best practices in data curation, such as self-describing entities, FAIR principles, and rich Hierarchical Event Descriptor (HED) annotations.
- Developed in conjunction with the BIDS Enhancement Proposal (BEP) 032 for micro-electrode neurophysiology data.

NeuroConv | *Lead Developer*

- Established an automated data conversion tool capable of reading more than 40 distinct data formats used by neurophysiology experiment devices in order to automatically write to the NeurodataWithoutBorders (NWB) standard.
- Designed universal APIs which transparently handled each layer of complexity to simplify the tasks of tagging, grouping, metadata transcription, temporal alignment, asset linking, buffering, chunking, and compression.
- Implemented a distributed cloud deployment system to run large-scale, off-site, batched conversions through Amazon Web Services (AWS).

NWB Inspector | *Lead Developer*

- Built a command line tool used by the NIH data archive to validate all data uploads, which ensures all submissions are provided automated suggestions for metadata improvements that enhance data findability and reuse.
- Mirrored the design, style, and functionality of linting tools such as flake8, pydocstyle, and ruff.

Highlighted Projects (continued)

Neurosift | *UX Consultant*

- Aided in the design of a Browser-based application for visualizing the contents of terabyte-scale neurophysiology datasets.
- Supports time series, images, videos, pose, tabular, annotated events, and all associated metadata through an intuitive exploration interface.
- Utilizes a specialized HDF5 web assembly implementation for increased efficiency when streaming data from the S3-backed archives.

NWB Benchmarks | *Core Developer*

- Managed a collection of unit and integration tests whose performance metrics (time, memory, and web packets) are sampled against a range of environmental run configurations, machine specifications, and bandwidth limits with the results used to assess the performance of HDF5 and Zarr file packaging parameterization against cloud streaming methods.
- Run outputs are aggregated into a common public database and built-in visualization tools generate reports of summary findings.
- Built as an extension on top of the popular Airspeed Velocity (ASV) benchmarking framework.

duct | *Developer*

- Maintained a simple command line tool for monitoring and maintaining records of process-specific system resource utilization during computational tasks.

Hierarchical Data Modeling Framework (HDMF) | *Contributor*

- Enhanced a generic schema language for defining structured data standards intended to be written to group-like back-end file types (such as HDF5 and Zarr) by generalizing, parallelizing, and optimizing lazy data buffering for terabyte-scale datasets.

NWB Widgets (Legacy) | *Developer*

- Drafted Jupyter widgets-based tools for visualizing the contents of terabyte-scale neurophysiology datasets.
- Supports many custom plugins for specific neural data types.
- Has some remote streaming capability, but is more intended for reading from local disks.
- Largely replaced by the Neurosift browser-based application.

Publications

Journal Articles

- 1 C. J. Gillon, **C. Baker**, et al., “Open data in neurophysiology: Advancements, solutions & challenges,” *eNeuro*, vol. 12, no. 11, 2025.  DOI: 10.1523/ENEURO.0486-24.2025.
- 2 K. Gunalan, R. Choudhury, et al., “DANDI: Shaping the future of neurophysiology and neuroimage data publishing and exploration through collaboration with the community, standards, and analytic platforms,” 2025, In preparation.
- 3 R. Ly, **C. Baker**, et al., “A practical evaluation of remote access methods for neurophysiology data in the cloud,” 2025, In preparation.
- 4 H. Mayorquin, **C. Baker**, et al., “NeuroConv: Streamlining neurophysiology data conversion to the NWB standard,” *Python in Science Conference, 2025*, 2025.  DOI: 10.25080/cehj4257.
- 5 J. Magland, J. Soules, et al., “Neurosift: DANDI exploration and NWB visualization in the browser,” *Journal of Open Source Software*, vol. 9, no. 97, p. 6590, 2024.
- 6 M. Hawrylycz, M. E. Martone, et al., “A guide to the BRAIN Initiative Cell Census Network data ecosystem,” *PLoS biology*, vol. 21, no. 6, e3002133, 2023.
- 7 **C. Baker**, E. Froudarakis, et al., “Inference of synaptic connectivity and external variability in neural microcircuits,” *Journal of Computational Neuroscience*, vol. 48, pp. 123–147, 2020.
- 8 **C. Baker**, V. Zhu, et al., “Nonlinear stimulus representations in neural circuits with approximate excitatory-inhibitory balance,” *PLoS computational biology*, vol. 16, no. 9, e1008192, 2020.
- 9 **C. Baker**, C. Ebsch, et al., “Correlated states in balanced neuronal networks,” *Physical Review E*, vol. 99, no. 5, p. 052 414, 2019.

Conference Presentations

- 1 **C. Baker**, G. Flynn, et al., “NWB GUIDE: Simplifying the conversion of neurophysiology data to NWB format,” in *Society for Neuroscience (SfN)*, 2024.
- 2 **C. Baker**, S. Weigl, et al., “NeuroConv: Automated conversion of neurophysiology data to NWB format,” in *Society for Neuroscience (SfN)*, 2023.
- 3 **C. Baker**, V. Zhu, et al., “Nonlinear computations in semi-balanced networks,” in *Computational and Systems Neuroscience (COSYNE)*, contributed talk (top 4% of abstracts), 2020.
- 4 **C. Baker** and R. Rosenbaum, “Inferring connectivity and latent input covariance from spike train correlations,” in *Computational and Systems Neuroscience (COSYNE)*, poster (top 43% of abstracts), 2018.

Workshop Sessions

- 1 **C. Baker**, R. Ly, et al., “Neurodata Without Borders and DANDI Archive Tutorial,” in *NWB and DANDI Workshop*, Baylor College of Medicine, 2024.
- 2 **Baker, Cody**, “Neuroconv and NWB GUIDE tutorial,” in *NWB User Days*, Janelia Research Campus, 2023.
- 3 **C. Baker** and R. Ly, “Neurodata Without Borders and DANDI Archive Tutorial,” in *Pre-conference*, COSYNE, 2023.
- 4 **C. Baker**, O. Ruebel, et al., “Neurodata Without Borders and DANDI Archive Hackathon,” in *NeuroDataShare*, Sainsbury Wellcome Centre, 2023.
- 5 **C. Baker** and G. Viejo, “Artificial intelligence, machine learning, computing, and visualization in neuroinformatics,” in *Open Data in Neuroscience (ODIN)*, Massachusetts Institute of Technology, 2023.
- 6 **Baker, Cody**, “NeuroConv tutorial,” in *NWB User Days*, Janelia Research Campus, 2022.